

- Q.25 Explain all types of elements used in PLC.
- Q.26 What are the limitations of electromagnetic relay?
- Q.27 What are the importance of DCS?
- Q.28 Draw and explain the block diagram of SCADA.
- Q.29 What are the latch and master controlled self-holding relays in PLC?
- Q.30 Write a short note on scan cycle with proper diagram.
- Q.31 Define RTU. What are the advantages of HMI?
- Q.32 Explain “Less” and “Less than equal” instructions in PLC.
- Q.33 Write down any five applications of PLC.
- Q.34 Draw and explain the block diagram of DCS.
- Q.35 What are the advantages of DCS in operation and safety?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write a short note of HMI. Write down any four applications of HMI. What are the differences between PLC and SCADA.
- Q.37 What is arithmetic instruction in PLC? Why it is used? Explain any two arithmetic instructions in PLC.
- Q.38 Draw and explain the block diagram of PLC. Discuss the memory structure of PLC. Write down any four applications of it.

No. of Printed Pages : 4 181562/121562/031562
Roll No.

6th Sem / IC, EI Sub : PLC, DCS and SCADA / Micro-Cont. & PLC Based Instr.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is not the building block of SCADA?
a) Auxiliary memory b) RTU
c) RTC d) CPU
- Q.2 Ladder diagram of AND gate is _____ combination of normally open relay contacts.
a) Series b) Parallel
c) Series and parallel both d) None of the above
- Q.3 Which one of the following is not involved in scan cycle?
a) Input read b) Output scan
c) Program execution d) None of the above
- Q.4 DCS stands for
a) Direct control system
b) Distributed control system
c) Digital control system
d) Distributed command system

- Q.5 _____ in PLC is a battery backed hardware component that tracks the actual time and date and allow control to perform time-based tasks.
- a) Watch dog timer b) Real time clock
c) Retentive timer d) HMI
- Q.6 DIV instruction is
- a) An arithmetic instruction
b) A timer instruction
c) A counter instruction
d) None of the above
- Q.7 LEQ is a/an _____ instruction in PLC.
- a) Arithmetic b) Counter
c) Timer d) Comparison
- Q.8 MTU stands for
- a) Master terminal unit b) Master time unit
c) Master task unit d) Major time unit
- Q.9 Which of the following is used for very large area in industries?
- a) SCADA b) PLC
c) HMI d) DCS
- Q.10 The device which is used to create, modify or troubleshoot a program in PLC is
- a) Memory b) Programming device
c) Power supply d) Processor

(2) 181562/121562/031562

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define PLC.
- Q.12 Expand RTC.
- Q.13 Write any one application of HMI.
- Q.14 SUB is a/an _____ instruction in PLC.
- Q.15 Expand RTU.
- Q.16 Define scan rate.
- Q.17 DDC refers to the automated control of a process by computers and microprocessors with sensors. (True/False)
- Q.18 MUL is an instruction in a PLC is a basic way to transfer data by replacing it from one location to another. (True/False)
- Q.19 Write any one manufacturer of PLC.
- Q.20 A PLC is the part of block diagram of SCADA. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write down any five difference between PLC and relays.
- Q.22 What are the different manufacturers of PLC?
- Q.23 Explain the working of DDC.
- Q.24 Write a short note on comparison instruction in PLC.

(3) 181562/121562/031562